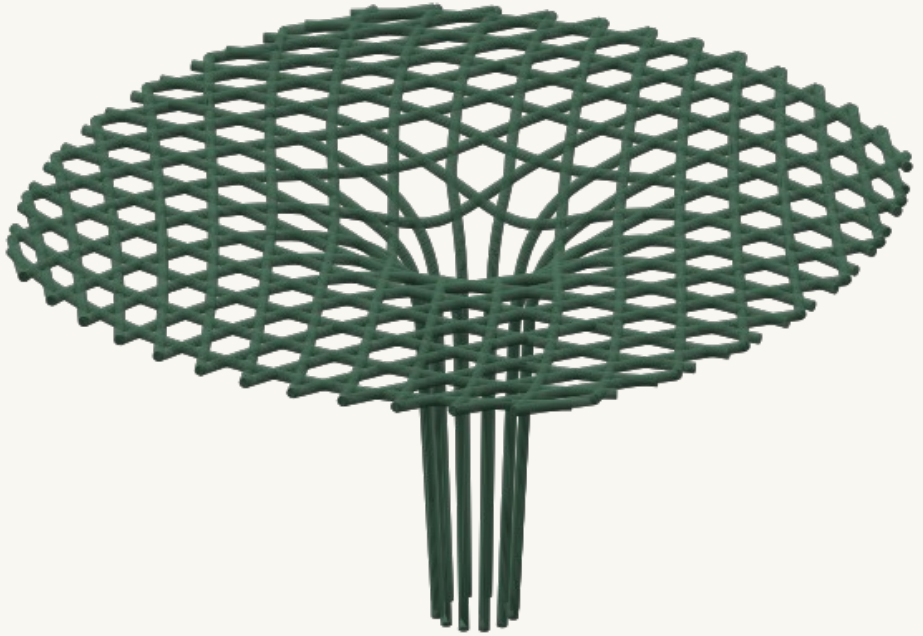
# VineWeave Canopy

A tree-like climbing-plant support for pedestrian shade

Parametric design · Planting · Shade performance



TEAM & RESPONSIBILITIES

**臧唯羽**

Form & parametric modelling

**林新淇**

Materials & construction

**虞寒哲**

Shade performance & application

**徐世程**

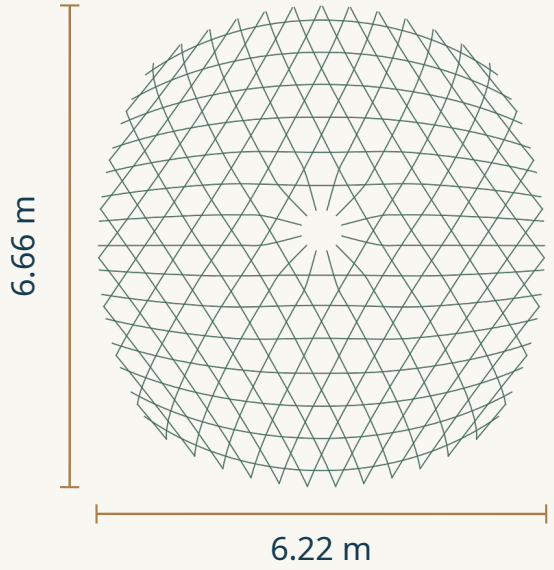
PPT production & integration

**陈淼**

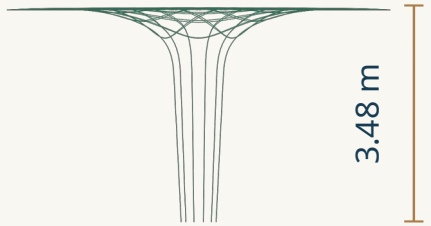
English presentation & Q&A

# Define the final geometry—and how it is measured

Dimensioned plan/elevation views; a final parameter schedule, separate from sensitivity settings.




**PLAN**




**ELEVATION**

Curved support inclination is defined by its endpoint chord, not a local tangent.

| FINAL PARAMETER | VALUE / DEFINITION |
|---|---|
| Overall height | ≈3.48 m |
| Plan extents | ≈6.22 × 6.66 m |
| Maximum horizontal span | ≈6.91 m, exterior mesh |
| Support inclination* | 45.9–50.7° to horizontal |
| Member exterior diameter | 80 mm |
| Corner / expansion | 0.85 / 1.08 |

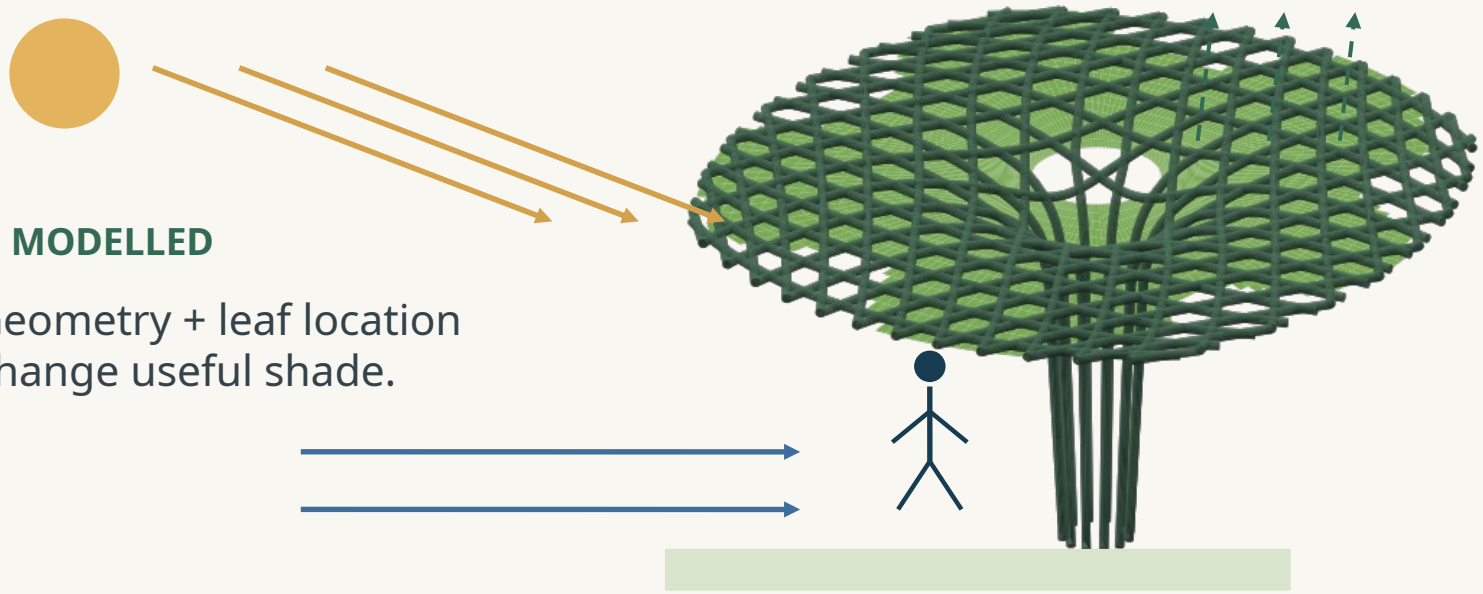
*Angle of each ground-reaching centre-line endpoint chord. Not a uniform member slope.

Model records P1 and supplemental geometric check. Centre-line views; exterior dimensions rounded. Drawing sheets accompany the PPT.

# How the canopy can improve pedestrian comfort

Three physical pathways, with the strength of evidence shown next to each.

## 01 DIRECT SUN



**MODELLED**

Geometry + leaf location change useful shade.

## 03 PLANT WATER

Evapotranspiration may exchange latent heat; it needs healthy, watered plants.

## 02 AIRFLOW

Open sides retain possible air paths.
Dense foliage can also add resistance.

### 01 SHADE → LOWER EXPOSURE

Direct sun on people and ground

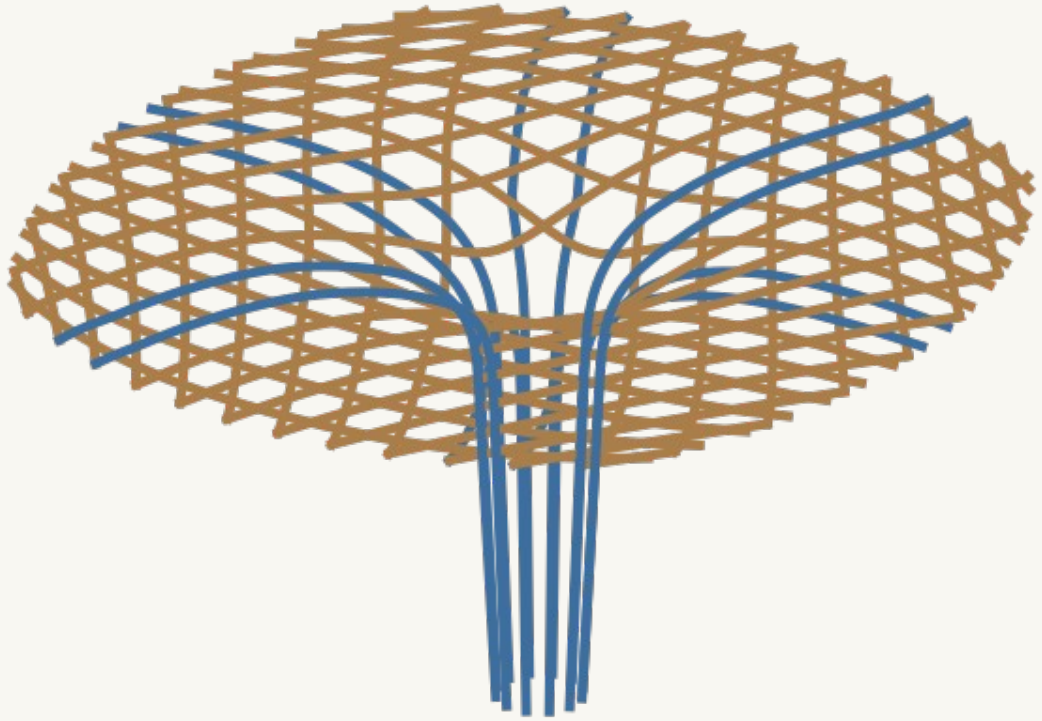
### 02 / 03: QUALITATIVE MECHANISMS

**Shade % does not equal cooling °C.**

Shade: project P2/P5. Greening mechanisms: [1,2]. Arrows are conceptual; no temperature or airflow field is simulated.

# Separate the support role from the crown role

Classify members by whether their centre lines reach the ground.



| ZONE | PROPOSED SECTION | LENGTH |
|---|---|---|
| 12 steel supports | Q355B tube Ø80 × 4 mm | 69.38 m |
| 47 bamboo crown members | Engineered bamboo solid Ø80 mm | 219.28 m |
| TOTAL | 59 members | 288.66 m |

Steel supports address compression and wind bending. The bamboo crown is governed by stiffness, treatment and replaceability.

Project take-off and proposed member schedule (P1, P3). Sections and material mass remain preliminary specifications.

# Five criteria guide the material selection

Durability · cost · thermal conductivity · carbon · environmental adaptability.

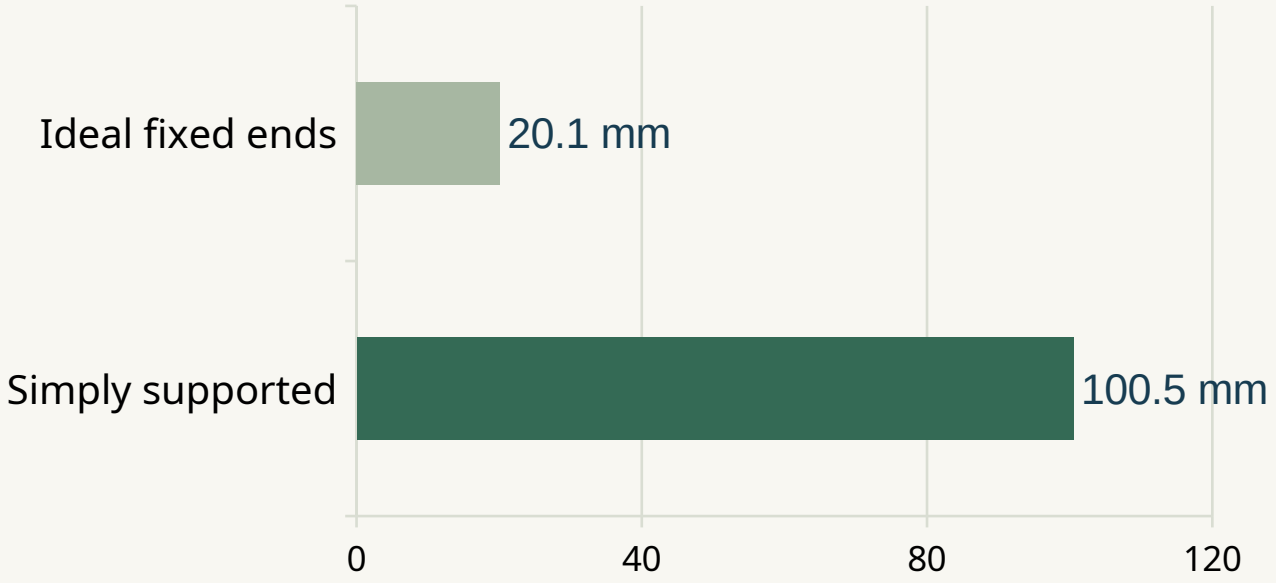
| MATERIAL / SECTION | BENDING* kN·m | λ* W/(m·K) | CARBON* $kgCO_2e/kg$ | MATERIAL COST* CNY / 288.66 m |
|---|---|---|---|---|
| Q355B tube Ø80×4 | 6.14 | 50 | 2.15 | 18,394 |
| Q355B tube Ø80×3 | 4.78 | 50 | 2.15 | 13,983 |
| Engineered bamboo Ø80 | 1.01 | 0.22 | 0.40 | 10,158 |
| Treated timber Ø80 | 0.85 | 0.14 | 1.10 | 6,530 |
| Raw bamboo Ø80 | 0.40 | 0.18 | 0.25 | 4,330 |

| DURABILITY | ENVIRONMENTAL ADAPTABILITY |
|---|---|
| Protect steel coatings; treat bamboo; detail inspection and replacement. | Check moisture, UV, wind exposure, local supply and plant suitability. |

Source: material-comparison workbook (P3). *Preliminary inputs; product, processing and factor boundaries need verification.

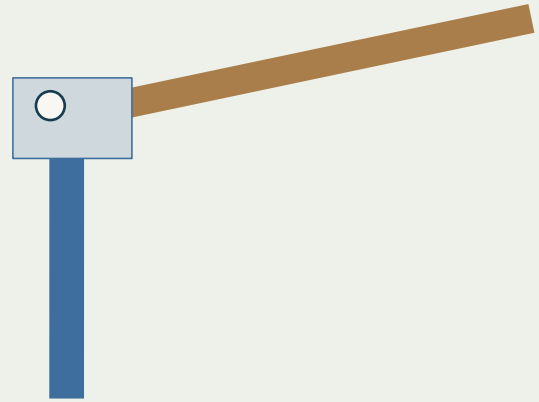
# The crown is stiffness-sensitive: joints matter

Load assumption → boundary condition → deflection → required connection behaviour.




**6.65 m SPAN · ADOPTED L/200 = 33.25 mm**

## PROPOSED SUPPORT-HEAD NODE



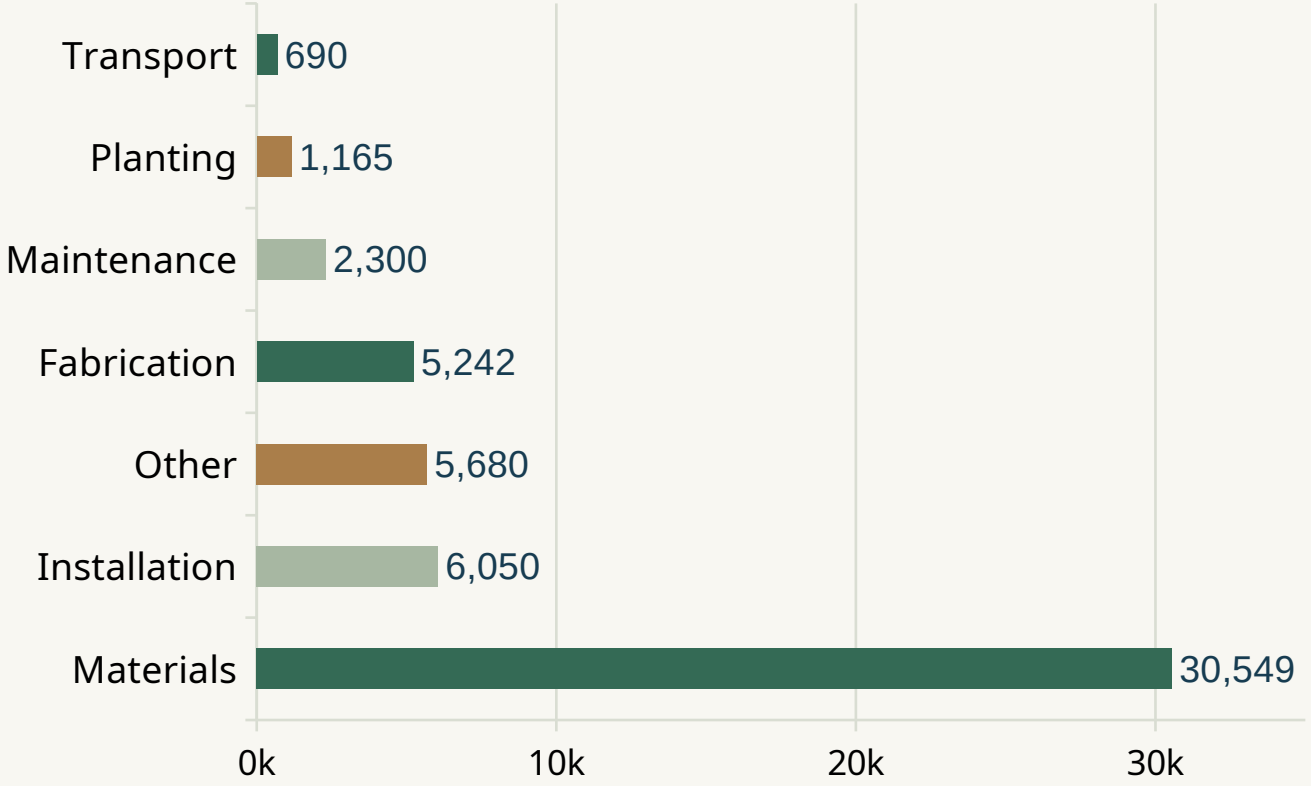
Node plate + sleeve + structural adhesive
with high-strength bolts

Concept only; rotational restraint must be verified.

Simplified beam calculation and proposed node method (P3). Wet foliage 18 kg/m² and bamboo properties are assumed inputs.

# The unit budget includes fabrication, installation and care

Preliminary total: CNY 51,676 per unit—not simply the price of the members.




**MATERIALS = 59.1% OF TOTAL**

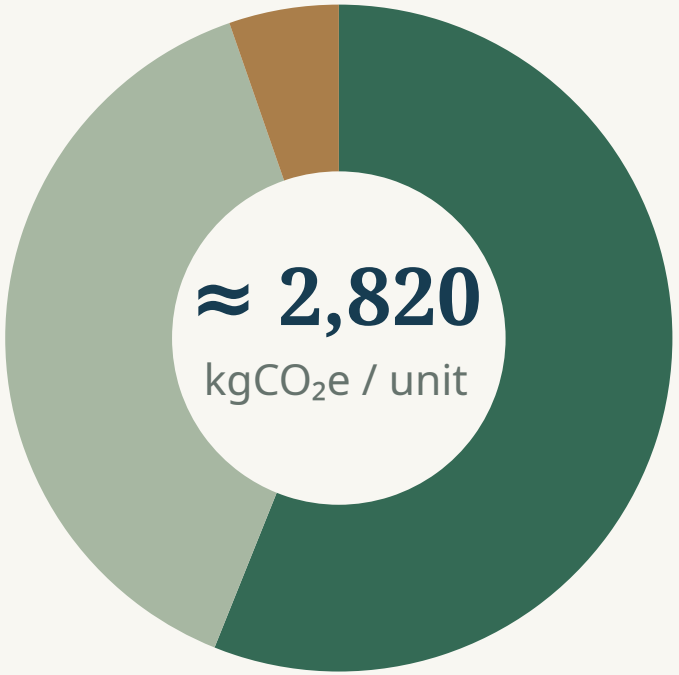
| KEY COMPONENT | CNY |
|---|---|
| Bamboo crown | 7,716 |
| Steel supports | 4,421 |
| 12 rigid-head nodes | 12,000 |
| Counterweight | 3,402 |

**These component rows are part of the full total. Nodes are a larger allowance than the primary steel members.**

Source: Cost-Analysis.xlsx (P3). All rates remain reference estimates rather than confirmed quotations.

# Account for the members, base and maintenance needs

Current carbon estimate: approximately 2,820 $kgCO_2e$ per unit.




| CONTRIBUTOR | $kgCO_2e$ |
|---|---|
| Primary members | 1,581 |
| Counterweight | 1,089 |
| Other components | 150 |

**Counterweight ≈ 39% of the current estimate**

| STEEL | BAMBOO | CONNECTIONS |
|---|---|---|
| Protect coating | Treat and inspect | Check fixings and grout |

Workbook estimate (P3). Factor boundaries and node inclusion require checking; no biogenic-carbon credit is taken.

# Keep the planting mix conditional on the site

Source proposal: 2 Boston ivy + 1 Rangoon creeper per unit; trumpet creeper is an alternative.

**PRIMARY • 2 PLANTS**

## Boston ivy

*Parthenocissus tricuspidata*

Deciduous: summer leaf cover, winter leaf loss. Adhesive tendrils; train and check grip on the support.

**COMPANION • 1 PLANT**

## Rangoon creeper

*Combretum indicum*

Warm-climate climber. Check winter survival before adoption; provide guides and routine tying.

**ALTERNATIVE**

## Trumpet creeper

*Campsis grandiflora**

Vigorous woody climber; check roots, pruning demand and attachment. Confirm nursery species.

**ROOTS + CARE**

Source allowance: 1.2 $m^3$ substrate, drip irrigation, training and pruning.

*“Trumpet creeper” can refer to different Campsis species/hybrids. Nursery identification remains provisional.

# Project shade onto one fixed reference zone

30°N · assumed X=east/Y=north · true solar time 09–15 · no surrounding obstructions.

| | |
|---|---|
| 01 | SUN DIRECTION |
| 02 | MODEL + LEAF ENVELOPE |
| 03 | GROUND PROJECTION |
| 04 | TARGET INTERSECTION |

## IDEAL GEOMETRIC OVERLAP

Continuous-leaf shadow ∩ target ÷ target area

## EQUIVALENT DIRECT-BEAM BLOCKAGE

$$A_{eq} = A_{bare} + (1 - \tau)(A_{leaf} - A_{overlap})$$

Divide Aeq by target area; τ = 0.15.

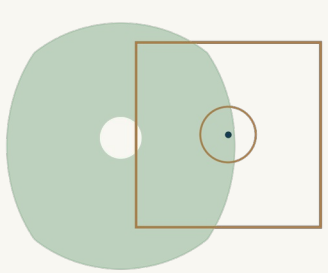
**TARGET: 5 × 5 m MINUS r = 0.75 m → 23.23 m²**

Not net usable floor area.

Solar geometry: NOAA [3]. Project calculations and scenario assumptions: P2, P5. Foliage/transmittance are not measured.

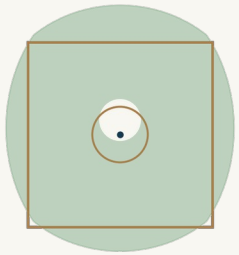
# The shadow position and solar season both matter

Ideal continuous-leaf overlap: a high noon value does not mean continuous occupied-hour shade.



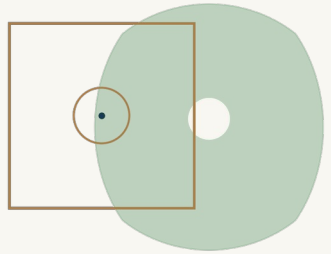
09:00

**47.2%**



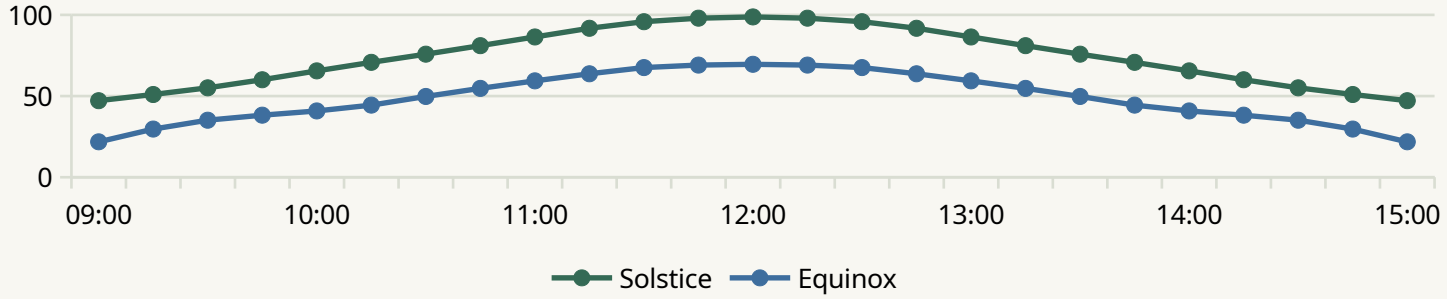
12:00

**98.7%**



15:00

**47.2%**



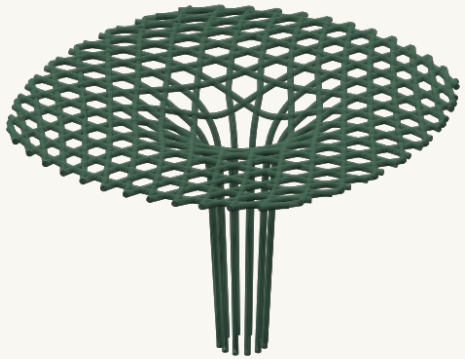

**09–15 TIME MEAN**

**Solstice 75.3%**
**Equinox 49.8%**

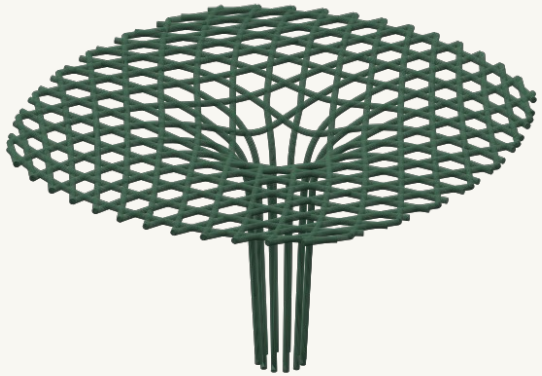
Project projection data (P2, P5). Solar-season comparison under the assumed foliage geometry.

# Seven models test three geometry controls

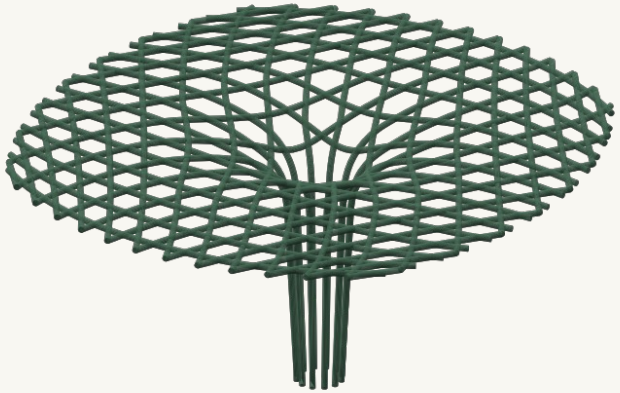
One variable at a time; same target, 50% plan cover, solstice solar hours and τ = 0.15.



SMALLER CROWN



BASELINE



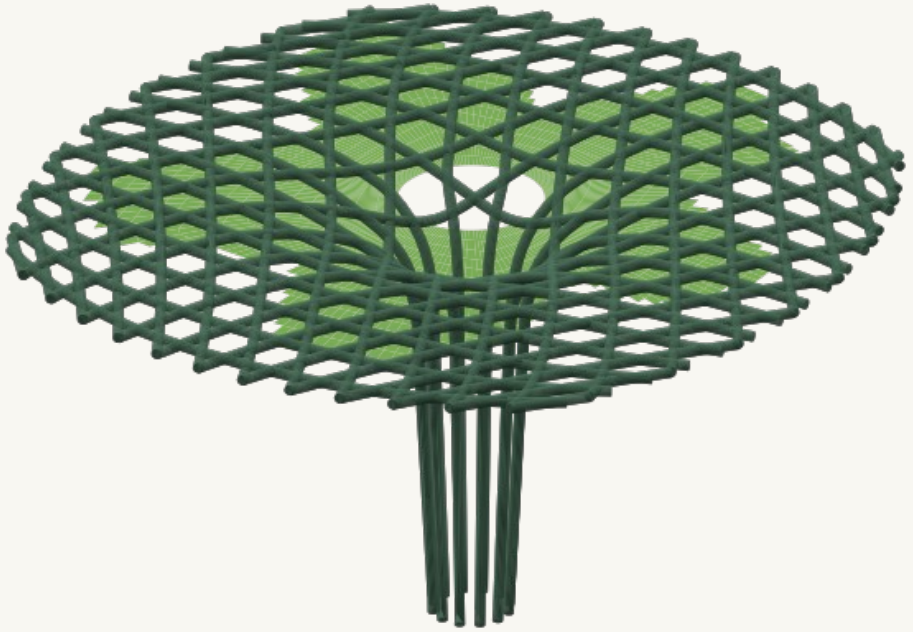
LARGER CROWN

| CONTROL | TESTED SETTINGS | MEAN BLOCKAGE: LOW / BASE / HIGH |
|---|---|---|
| Upper expansion | 0.92 / 1.08 / 1.24 | 52.3% / 60.2% / 66.1% |
| Upper-height control | −0.4 / 0 / +0.4 m | 62.6% / 60.2% / 57.8% |
| Corner factor | 0.55 / 0.85 / 1.00 | 61.8% / 60.2% / 59.1% |

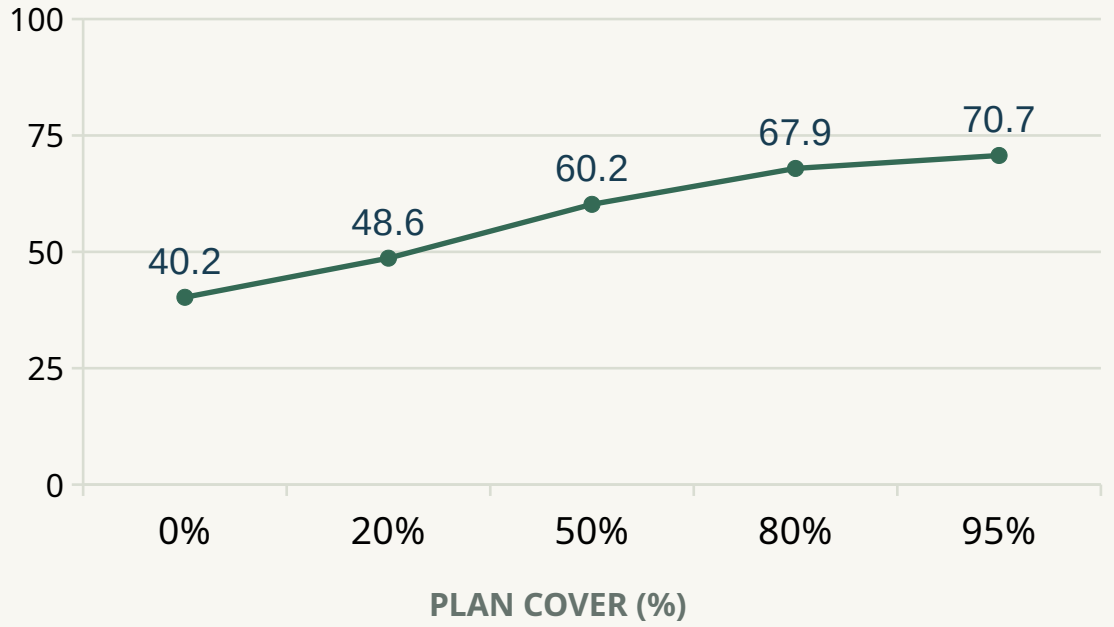
Project parameter study (P5). Controls are not uniform scaling; larger cover percentage does not imply equal plant/material input.

# Coverage amount alone does not determine useful shade

Fixed final structure; five spatial-cover scenarios with the same optical assumptions.



STATIC VIEW: 50% PLAN COVER • ANIMATION IN PPT

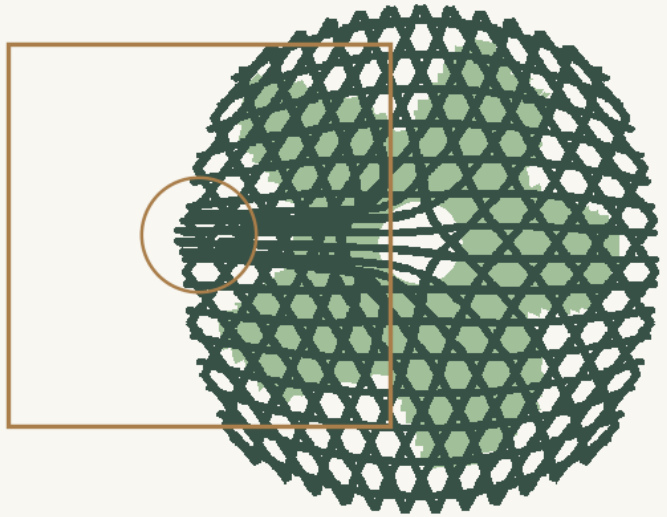



**Same 50% cover, different layouts at noon: 77.5% / 71.4% / 62.6% blockage.**

Project spatial-cover calculation (P5). Frame/leaf overlap deducted, τ = 0.15; equivalent direct-beam blockage, not cooling.

# Put occupied-hour shade into the layout decision

A rest node must coordinate seating, through movement, roots and a substantial base.



STATIC VIEW: 15:00 • FIXED 50% PLAN COVER

## 01 SEATING LOCATION

Use the occupied-hour footprint, not just the crown outline or shadow centroid.

## 02 SPACE FOR ACCESS & ROOTS

Allow for circulation, counterweight, root zone, irrigation and maintenance.

## 03 VALIDATE THE OUTCOME

Confirm site and planting; compare radiation, temperature, humidity and wind.

Conceptual application P4; fixed-50%-cover projection P5. No real-site compliance or measured comfort result is claimed; method lead [4].

# Coordinate prefabricated modules and site interfaces

Numbered components help assembly; long members, heavy base and grouted joints constrain it.

| MODULE | WHAT IS PREPARED |
|---|---|
| Crown | 47 formed/treatment-ready bamboo members |
| Supports | 12 proposed galvanised steel members |
| Connections | Head nodes, sleeves, clamps and base flange |
| Foundation | Precast counterweight with anchorage |
| Planting | 2 + 1 plants, substrate and drip provision |

01 **Prefab, treat & number**

02 **Lift base and set out**

03 **Support, align & assemble**

04 **Grout, cure & recheck**

05 **Plant, train & restore**

**14 person-days estimated ≠ 14 calendar days. The ≈9.07 t base controls lifting; grouted nodes limit lossless reversal.**

Project member/assembly and cost schedule (P3, P4). Proposed detail and labour quantities, not a tested construction programme.

# Connect geometry, construction and useful shade

Three design workstreams now have traceable outputs.

GEOMETRY

## Rhino/GH model and parameters

Final form, drawings and material take-off

MATERIALS & BUILD-UP

## Screening, sections, budget and modules

Proposed interfaces and implementation constraints

PERFORMANCE

## Time-dependent shade and sensitivity

Geometry, cover and spatial-distribution effects

**7 PARAMETER MODELS**

**5 COVER STATES**

**1,750 TIME-STEP RESULTS**

**Next: confirm products and joints, check plant suitability, and test the real site.**

Synthesis of project records P1–P5. Product, joint and site-specific evidence are the next verification priorities.

# References and project evidence

Method sources and project inputs are separate; drawings, brief and schedules accompany the deck.

**[1] OUTDOOR SHADE & COMFORT**

Watanabe et al. (2014). Evaluation of outdoor thermal comfort in sunlight, building shade, and pergola shade during summer in a humid subtropical region. Building and Environment, 82, 556–565.

**[2] GREENING MECHANISMS**

Hoelscher et al. (2016). Quantifying cooling effects of facade greening: Shading, transpiration and insulation. Energy and Buildings, 114, 283–290.

**[3] SOLAR GEOMETRY**

NOAA Global Monitoring Division. General Solar Position Calculations.
gml.noaa.gov/grad/solcalc/solareqns.PDF

**[4] COMFORT METHOD**

UTCI Calculator / COST Action 730.
www.utci.org/utci_calc.php
Method lead only; no project UTCI result is reported.

**PROJECT RECORDS**

P1 Model & take-off • P2 Ideal shade analysis • P3 Cost-Analysis.xlsx
P4 Proposed design & build-up • P5 Parameter and spatial-cover simulation

Project models, optical scenarios and estimates are identified separately from published literature.